LECTURE

Retrieval-Augmented Generation

How language models answer with knowledge they were never trained on

Why RAG exists

Stale knowledge

Models know only what was in their training data, up to a cutoff date.

No private data

Your documents, tickets and wikis were never part of training.

Hallucination

Without evidence, a model may produce fluent but wrong answers.

The RAG pipeline

1

Ingest

Split documents into chunks and embed them.

2

Retrieve

Embed the question and find the nearest chunks.

3

Augment

Place the chunks in the prompt with the question.

4

Generate

The model answers and cites its sources.

Chunks and embeddings

Chunking

Aim for self-contained passages of a few hundred tokens, with a small overlap. Split on headings and paragraphs, not mid-sentence.

Embeddings

A model maps text to a vector so that similar meaning lands close together. Search becomes a nearest-neighbour lookup in a vector index.

Improving retrieval

Technique	What it does
Hybrid search	Combines keyword (BM25) and vector scores
Reranking	A stronger model reorders the top candidates
Query rewriting	Turns a vague question into a better search query
Metadata filters	Restricts search by date, source or permissions

Evaluating and debugging

Retrieval recall

Was the right passage in the top results?

Faithfulness

Does the answer stay within the retrieved text?

Answer relevance

Does it actually address the question asked?

Common failure

Relevant text exists but is split across chunks.

TAKEAWAY

RAG gives a model an open book instead of a good memory.

Good chunks, good retrieval, grounded answers.